Youchul Jeong

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Education

University of the Aegean, Department of Marine Science — Mytilene, Greece
Ph.D. Candidate in Marine Science — *Oct 2025 - Present*
- Supervisor: Assoc. Prof. Konstantinos Topouzelis
- Marine Remote Sensing Group (MRSG) — *May 2025 - Present*

Pusan National University, Department of Earth Environmental System — Busan, South Korea
Master of Science in Oceanography — ***Feb 2024***
- MSc Thesis "A Study on Monitoring Floating Marine Macro Litter using a Multi-spectral Sensor and Classification based on Deep Learning." (Supervisor: Prof. Young-Heon Jo)
- Oral presentation award, Ocean & Atmosphere Science Interdisciplinary Studies Conference
- BrainKorea21 Scholarship (Earth Environmental Systems Education Research Group)

Pusan National University, Department of Oceanography — Busan, South Korea
Bachelor of Science in Oceanography — ***Feb 2022***
- BSc graduation paper "A Study on Image-Based Marine Litter Detection Techniques." (Supervisor: Prof. Young-Heon Jo)
- Center of Remote Sensing Lab — *Jan 2021 - Apr 2025*
- Academic Excellence Scholarship from the Department of Oceanography

Research Appointments

Scientific Researcher — *May - Oct 2025*
Department of Marine Science, University of the Aegean — Mytilene, Greece
- Floating marine debris research work using drone and satellite imagery in the Marine Remote Sensing Group (MRSG)

Research Assistant — *Sep 2022 - Dec 2022; Sep 2023 - Apr 2025*
Marine Research Institute, Pusan National University — Busan, South Korea
- Conducted drone fieldwork and developing a deep learning model to detect marine debris in the ocean.
- Developed innovative methodologies including drones, AI, and multi-spectral sensors to contribute to published articles on remote sensing.

Research Experience

Plastic Litter Project 2026 (PLP2026) — *Jun - Aug 2026*
- Led the design and field deployment of 18–90 m^2 floating plastic-litter targets in the Gulf of Gera, Lesvos, coordinating their construction, monitoring, and maintenance.
- Led the processing and analysis of satellite and drone imagery to assess floating plastic detection across multiple spatial and spectral scales.

Plastic Litter Project 2025 (PLP2025) *May 2025*

- Constructed floating plastic-litter targets using HDPE tarps and wooden planks in 1 m^2, 2 m^2, and 3 m^2 configurations for the PLP2025 field experiment.
- Assisted with field deployment, monitoring, repair, and retrieval of six target units in the Gulf of Gera, Lesvos.

Drone Analysis of Beach Safety and Beachgoer Behavior *Nov 2023, Feb - Aug 2024*

- Orchestrated drone flights and implemented the YOLOv8 algorithm to observe beach safety.
- Conducted surveys of beachgoer population to understand motivations and improve beach conditions.

Classification and Monitoring of Floating Marine Macro Litter (MSc Thesis) *Jun 2022 - Dec 2023*

- Utilized drones with multispectral sensors to create a custom FMML dataset.
- Implemented deep learning models for efficient litter classification and continuous monitoring.
- Incorporated drone atmospheric correction using DROACOR in collaboration with Dr. Daniel Schläpfer (ReSe Application LLC).
- Worked with advisors to promote conceptual framework and fieldwork at international conferences.

Floating Marine Debris and Physical Ocean Phenomena in S-ORS *April 2021 - Dec 2023*

- Applied drone technology and Applied Drone Automatic Flight Advisory in open ocean conditions.
- Developed a CNN model for detecting marine debris for Socheongcho Ocean Research Station.

Drone Survey of Spotted Seal Population *May 2022 - Nov 2023*

- Conducted drone surveys to assess spotted seal population on Baekryeong Island and gather high-resolution imagery, contributing crucial data to understanding spotted seal dynamics.

Floating Marine Litter, Korea Institute of Ocean Science & Technology *May - Oct 2023*

- Worked with Dr. Won Joon Shim (KIOST) to conduct on-site surveys of microplastics and floating marine litter.
- Compared and analyzed results of drone and multispectral sensor surveys.

Coastal Topography and Seabed Survey *Jul - Aug 2023*

- Conducted RTK-GPS and drone surveys to analyze coastal sand loss and seabed depth change.

UAV Team Leader, Korean Air Force *May 2016 - May 2018*

- Assembled military training UAVs, managed GPS element procurement, and led team activities.

Teaching Experience

Marine Debris in Circular Blue Economy [SBE3012] *Dec 18, 2025*

- Taught and presented “Marine litter detection” at Universiti Malaysia Terengganu as part of SustainaBlue HEIs in Malaysia and Indonesia.

Interpretation of marine data [Course No. MS3200990] *1st semester 2023*

- Led fieldwork for 40 undergraduate students to acquire Sargassum multi-spectral images.
- Taught data processing using MATLAB as part of the project curriculum.

Coastal Practices Survey [Course No. MS3200295] *1st semester 2022, 2023*

- Delivered theoretical lectures on YOLOv5, covering the installation process and algorithms.

Peer-Reviewed Publications

Youchul Jeong, Jisun Shin, Jong-Seok Lee, Ji-Yeon Baek, Daniel Schläpfer, Sin-Young Kim, Jin-Yong Jeong, Young-Heon Jo. "A Study on Monitoring Floating Marine Macro-Litter using a Multi-Spectral Sensor and Classification based on Deep Learning." Remote Sensing, 2024.

Sin-Young Kim, Jong-Seok Lee, **Youchul Jeong**, Young-Heon Jo. "Decomposition of Submesoscale Ocean Wave and Current Derived from UAV-Based Observation." Remote Sensing, 2024.

Conference Presentations

Youchul Jeong et al., "A Study on Monitoring Floating Marine Macro Litter using a Multi-spectral Sensor and Classification based on Deep Learning." Poster presentation at 11th Asian Workshop on Ocean Color, Nagoya, Japan, 2023.

Youchul Jeong et al., "A Study on Monitoring Floating Marine Macro Litter using Multi-spectral Sensor and Classification based on Deep Learning." Oral presentation at Korea Ocean Research Conference, Seoul, South Korea, 2023.

Youchul Jeong et al., "A Study on Monitoring Floating Marine Macro Litter using a Multi-spectral Sensor and Classification based on Deep Learning." Oral presentation at Ocean & Atmosphere Science Interdisciplinary Studies Conference, Busan, South Korea, 2023.

Youchul Jeong et al., "A Study on Classification and Monitoring of Floating Marine Macro Litter using Multispectral Imagery in UAV and Deep Learning." Poster presentation at European Geoscience Union General Assembly, Vienna, Austria, 2023.

Youchul Jeong et al., "A Study on Classification and Monitoring of Floating Marine Macro Litter using Multispectral Imagery in UAV and Deep Learning." Oral presentation at 10th Asian Workshop on Ocean Color, Indonesia, 2022.

Technical Skills

- Programming and analysis: MATLAB, Python
- Remote sensing software: ENVI 5.2, DROACOR, ACOLITE
- Satellite imagery: PRISMA, EnMAP
- UAV operations: 100+ research flights; South Korean UAV license for 25-150 kg aircraft
- Languages: Korean (native), English (proficient), Greek (beginner)